

Global CO2 Emissions Tracker

Analyzing Historical Climate Trends & Decarbonization Trajectories (1990-2020)

A Project by **Krisna Riyadi**

PYTHON / PANDAS

LOOKER STUDIO

PROJECT DISCLAIMER



Notice for Portfolio Reviewers & Recruiters

This is a personal project created solely for educational and portfolio purposes. The data used is a public dataset obtained from World Bank Open Data and reflects only a specific historical period. This project is not affiliated with, sponsored by, or representative of the internal data, guidelines, or official views of any related companies.

BACKGROUND & SELECTION

The Climate Challenge

Climate change is one of the most pressing global emergencies of our era. Greenhouse gases, particularly Carbon Dioxide (CO₂), act as the primary drivers of global warming.

Tracking the historical emissions of different nations is crucial for understanding climate trajectories and validating international environmental milestones like the Kyoto Protocol and the Paris Agreement.

Selection Considerations

- **Public Access:** Only publicly accessible datasets containing zero private or confidential metrics were utilized.
- **Global Relevance:** Focused on an industry topic that is widely recognized and easy to grasp for general audiences.
- **Data Credibility:** Sourced from top-tier, trusted organizations to ensure the underlying analytical validity can be fully justified.

BUSINESS PROBLEM & GOALS



The Challenge

Environmental NGOs, policymakers, and green-tech investors often struggle to allocate capital effectively because raw metrics are highly fragmented and difficult to interpret.



Identify Polluters

Determine exactly which countries are the primary contributors to global greenhouse gases, allowing focus areas to isolate the largest sources of emissions.



Track Key Trends

Quantify temporal shifts to reveal long-term regional patterns and analyze how environmental policy compliance alters carbon trajectories over time.

DATASET & TECH STACK

World Bank Open Data Ingestion

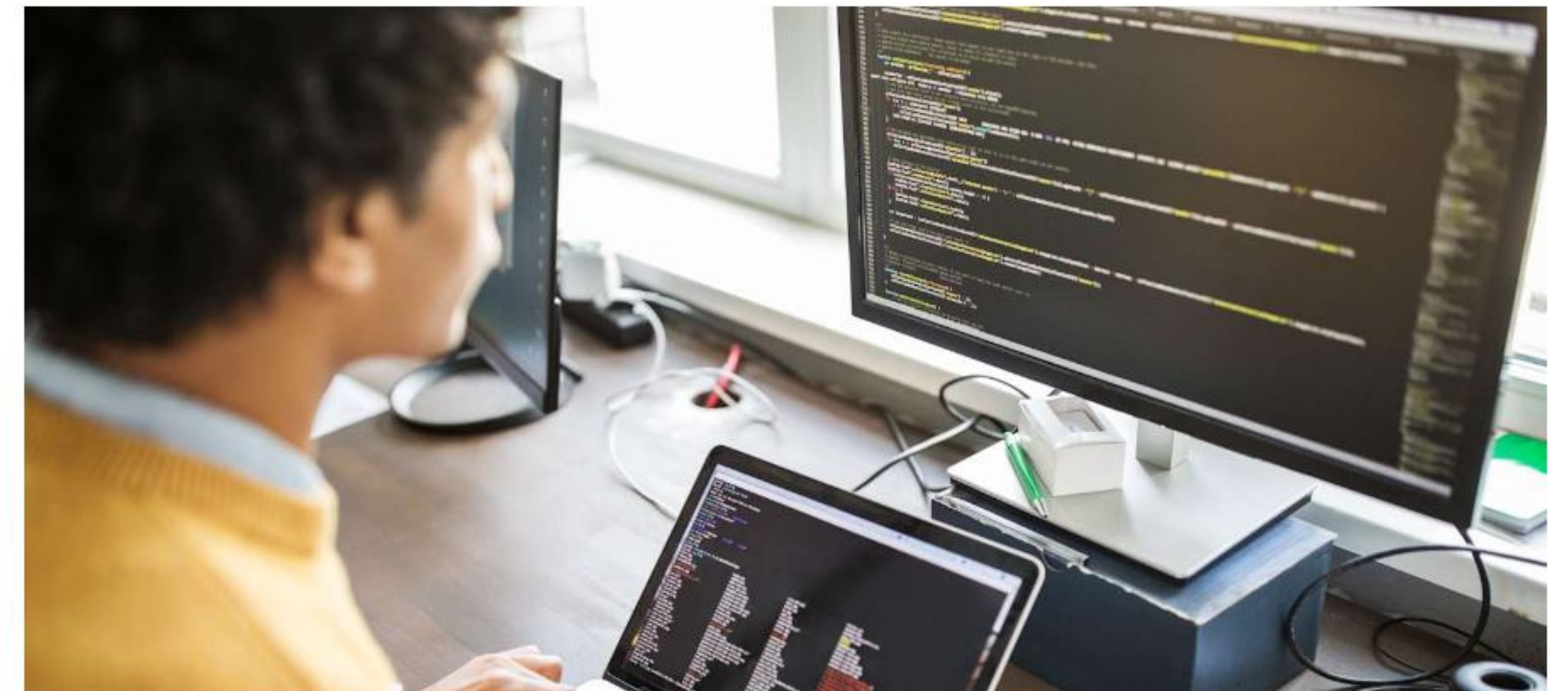
- **CO2 Emissions Excluding LULUCF:** Total historical outputs (kt) originating from all industrial, energy, and household sources.
- **CO2 Net Fluxes from LULUCF:** Dynamic land use, land-use change, and forestry offsets to track true carbon balance.

Google Sheets

Python (Pandas)

Looker Studio

Jupyter / Colab



ANALYTICAL METHODOLOGY



1. Ask & Prepare

Define business problems, stakeholder goals, and assess data integrity from Kaggle/World Bank.



2. Process

Exclusion of macro aggregates, reshaping data using Python, and missing value imputation.



3. Analyze

Executing calculations, highlighting geographic densities, and parsing national correlations.



4. Share & Act

Building interactive Looker Studio visualizations and creating strategic policy recommendations.

DATA PROCESSING PIPELINE

Extraction & Structural Review

Imported raw CSV structures directly into spreadsheet engines for baseline data exploration.

Excluding Aggregates: Aggressively filtered out macro-regional records (such as "Arab World" and "Euro area") to guarantee final counts isolate sovereign nations only, completely eliminating double-counting errors.

Cleaning & Reshaping (Pandas)

Unpivoting Formats: Reshaped the wide multi-year structure into a clean database format with six main dimensions using `pd.melt()`.

Imputation Mechanics: Resolved temporal data gaps via historical mean imputation (`df.fillna()`) to preserve time-series logic prior to reporting ingestion.

| GLOBAL KEY FINDINGS

Historical Insight Extraction

Dominant Contributors: Historically, China and the United States remain the primary drivers of global carbon emission outputs.

Upward Trajectory: Long-term tracking reveals persistent increases in carbon output, with massive, temporary contractions occurring in 2009 and 2020 during global economic contractions.

Economic Drivers: Analysis proves strong positive correlations between high national income brackets and overall absolute carbon emissions.



STRATEGIC ACTION PLAN



Target Large-Scale Polluters

Focus diplomatic and financial efforts on securing binding carbon-reduction commitments from leading historical drivers (USA & China) and regional hot spots (Europe).



Incentivize "Green Growth"

Introduce robust corporate and state incentives to decouple national GDP growth from high-carbon intensity production methods globally.



Replicate Systemic Reductions

Study historical drops (2009 & 2020) to structuralize planned, sustainable energy efficiencies within transport pipelines without relying on macro crises.



Enforce Strict ESG Criteria

Direct venture capital away from fossil fuels, while modernizing grid infrastructure to transition technology data hubs entirely to clean energy sources.

LET'S CONNECT!

Hi, I'm **Krisna Riyadi**

I am a highly analytical Data Professional specializing in setting up robust automated data pipelines, processing complex raw databases, and structuring interactive business dashboards that drive tangible strategies.

I build reliable data solutions to bridge raw system structures and critical executive insights. Let's discuss open opportunities or upcoming data initiatives.



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Thank You!

Open for Data Analyst Roles & System Strategy Discussions

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